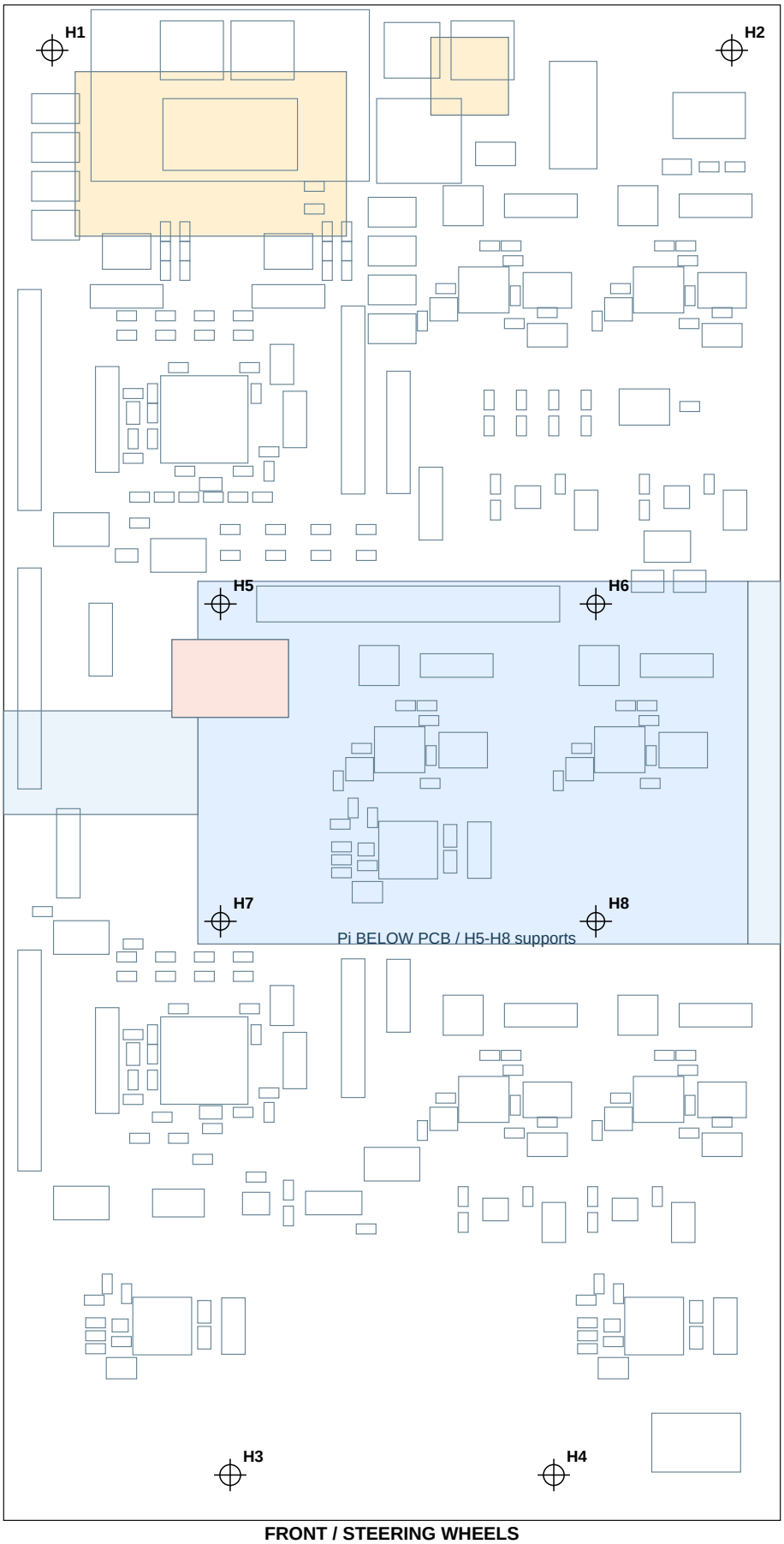
Lab 4 - provisional PCB fit template

Print at 100% / Actual size. Check the 50 mm bar before comparing with the car.

MEASURED HOLE CENTERS - PLACEMENT PREVIEW - NOT FOR FABRICATION

REAR / DRIVE MOTORS (0,0) at upper-left; x right, y down



FRONT / STEERING WHEELS

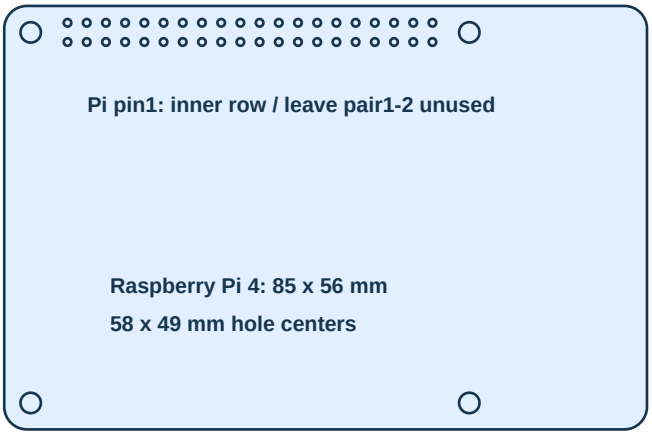
H1-H4: measured centers; M3 PCB drills proposed. Chassis holes measured 4.5 mm.

Check wires, steering travel, Pi below the PCB, port access and separate battery location.

This bar must measure 50 mm.

Pi alignment and vertical fit check

Top view below is 1:1. Place the unpowered Pi component-side-up to compare holes and pins.



USB / Ethernet face right.

GPIO row faces the REAR.

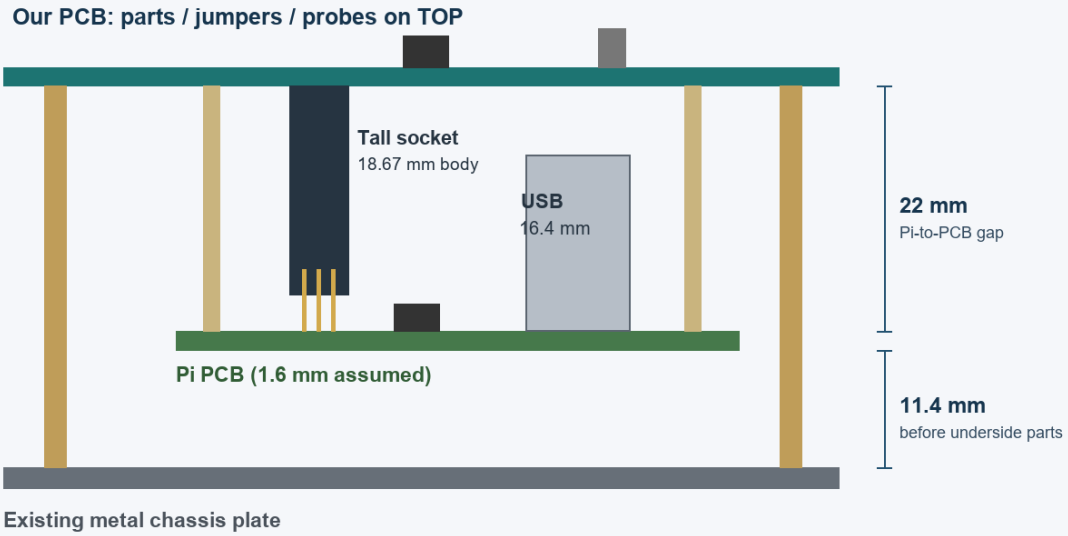
Socket mates Pi pins3-38 only.

Do not use this as a drill guide.

Main-PCB Pi origin: (30,89) mm.

How the boards stack - proposed dimensions

Pi component side faces UP. J70 socket is soldered to the UNDERSIDE of our PCB.



35 mm chassis standoffs proposed. Caliper heights recorded; final mating is checked at assembly.

Recorded geometry and remaining fit checks:

Chassis spacers: 35 mm nominal; four-hole paper fit passed with ~50 mm spacers.

Pi-to-PCB gap: 22 mm nominal; calculated socket engagement 5.17 mm.

Reported underside parts <5 mm; calculated nominal metal clearance 6.4 mm.

USB top measured ~18 mm from Pi board bottom; check any added cooler.

Working chassis pattern: final common CAD/physical holes still need resolution.

Still check unknown plug/cable/steering envelopes and assembled socket fit.

Pi hardware OD <=5.5 mm. All holes: 3 mm from hole edge. Socket first pair aligns to Pi3/4, not1/2.